Tesla Integrity Bot

1 **Faster answers**

Seconds, not a search through manuals

2 **Consistent guidance**

Same policy text behind every answer

3 **Auditable by design**

Every answer links to its source

AI CONSULTING PROPOSAL | TESLA MANAGEMENT

Tesla Integrity Bot

An AI assistant that answers ethics and compliance questions from approved policy documents, with sources shown.

BUSINESS BENEFITS

1 Faster answers

Seconds, not a search through manuals

2 Consistent guidance

Same policy text behind every answer

3 Auditable by design

Every answer links to its source

Demo scope: 5 synthetic policy documents | 250 indexed chunks | 15-question benchmark

PROBLEM STATEMENT

Scattered Documents, Difficult-to-Find Answers

KEY CHALLENGES

1. **Scattered documents**
   Ethics, privacy, security and confidentiality rules sit in separate manuals.

2. **Hard to search**
   Keyword search misses answers when staff use different wording.

3. **Inconsistent reading**
   Different teams interpret the same rule differently.

4. **Unverifiable AI answers**
   Generic chatbots answer fluently without citing company policy.

BUSINESS IMPACT

Lost productive time
Skilled staff searching instead of building.

Compliance exposure
A misread rule can become a violation.

Repeat questions
Ethics teams re-answer routine queries.

Shadow AI risk
Staff turn to unvetted external tools.

Ask in plain English. Get a cited answer.

1. **Ask naturally**
   No legal titles or section numbers needed.

2. **Search by meaning**
   Finds the most relevant policy passages.

3. **Answer from evidence only**
   Model must use retrieved text and decline otherwise.

4. **Show the sources**
   Document ID and excerpts shown beside the answer.

REAL DEMO OUTPUT | SYNTHETIC POLICY

EMPLOYEE QUESTION

What is the statutory SLA for processing customer vehicle telemetry deletion requests?

BOT ANSWER

Timeframe: process, execute and verify a complete deletion request within 30 business days of receipt.

Scope: all stored vehicle telemetry logs and associated cloud metadata (GDPR, CCPA, CPRA).

(Cited: TSLA-ETH-2026-02, Section 5.2, 30-Day Statutory Deletion SLA)

SOURCE SHOWN IN THE UI

TSLA-ETH-2026-02 (customer_data_privacy.txt) | Section: Full Document

USERS & WORKFLOW

Four user groups, one simple workflow

Employees

Get policy guidance in seconds

Metric: lookup time, satisfaction

HR & Ethics teams

Fewer repetitive questions

Metric: inquiry ticket volume

Compliance teams

More consistent awareness

Metric: audit vs. expert answers

People managers

Confident team decisions

Metric: manager feedback

EMPLOYEE WORKFLOW

1 Sign in	2 Ask	3 Retrieve	4 Answer	5 Verify
Mock login in demo; SSO planned	Type a question in plain English	Top 3 matching policy chunks	Grounded, cited response	Open the source excerpts

Implemented: Streamlit chat UI, source badges, chunk inspector. **Planned:** single sign-on, role-based access, audit logs.

WHY RAG?

RAG keeps answers grounded and policies easy to update

Approach	Plain-language questions	Grounded in company policy	Shows sources	Updates without retraining
Manual search	No	Yes	Yes	Yes
Keyword search	No	Yes	Yes	Yes
General LLM only	Yes	No	No	No
Fine-tuned LLM	Yes	Partly	No	No
RAG assistant	**Yes**	**Yes**	**Yes**	**Yes**

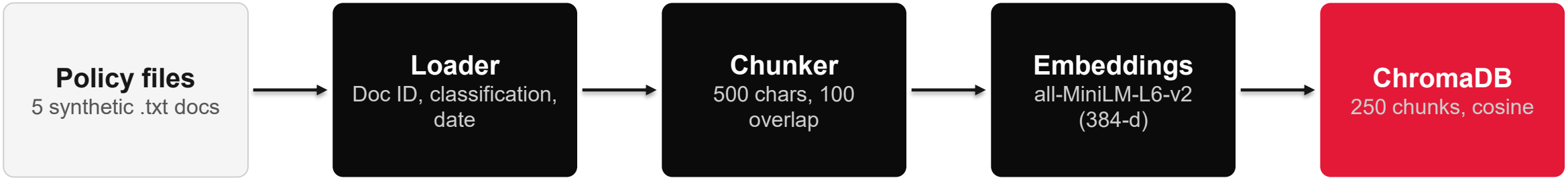
Trade-off: RAG quality depends on retrieval and document coverage. Qualitative assessment.

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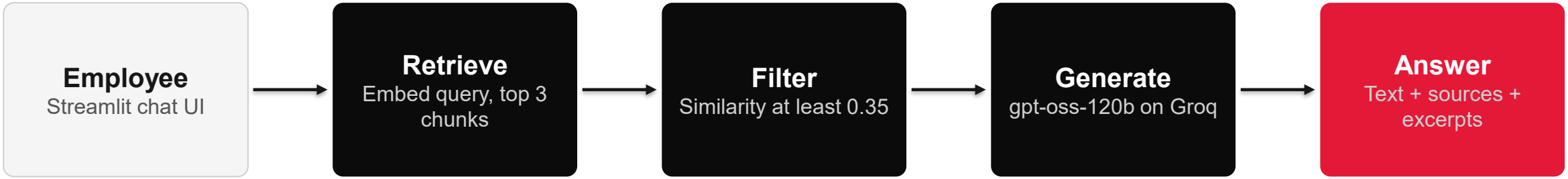
ARCHITECTURE

End-to-end RAG architecture

A. OFFLINE: DOCUMENT PREPARATION



B. REAL TIME: QUESTION ANSWERING



Guardrails in the prompt: answer only from retrieved context, no invented policies, cite document and section, say so when information is missing. **Temperature 0.0 by default.**

LLM VS. SLM

LLM in the demo. SLM to be tested in the pilot

SMALL MODEL (SLM)

e.g. 3B-8B, can run on-premise

- Lower compute cost
- Private / air-gapped option
- Needs testing on policy nuance

NOT TESTED IN DEMO

LARGE MODEL (LLM)

e.g. 70B-120B, used in demo

- Stronger synthesis of multi-part answers
- Cloud API: per-token cost
- Policy text leaves the network

USED IN DEMO

MODELS IN THE DEMO

Embedding
all-MiniLM-L6-v2
Small, local, 384-d vectors

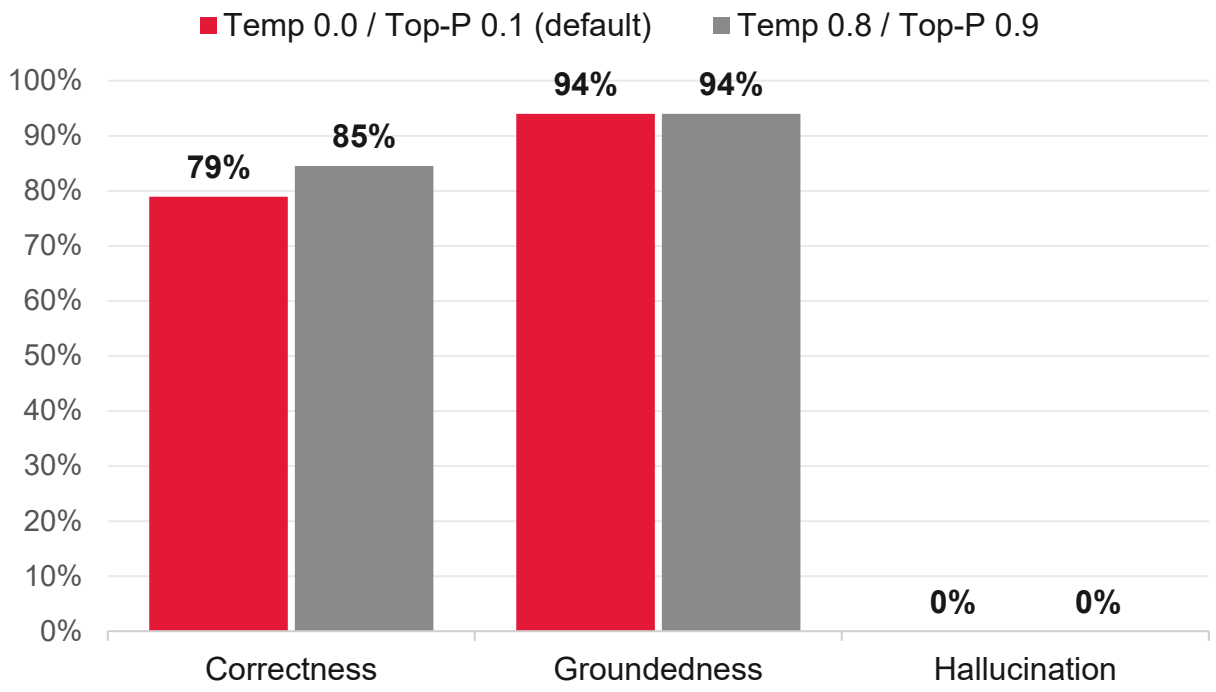
Generation
openai/gpt-oss-120b
Via Groq Cloud API

Recommendation: keep the LLM for the pilot, and benchmark one SLM on the same questions and retrieved context. Choose on accuracy, latency, cost and data-privacy needs, not on size alone.

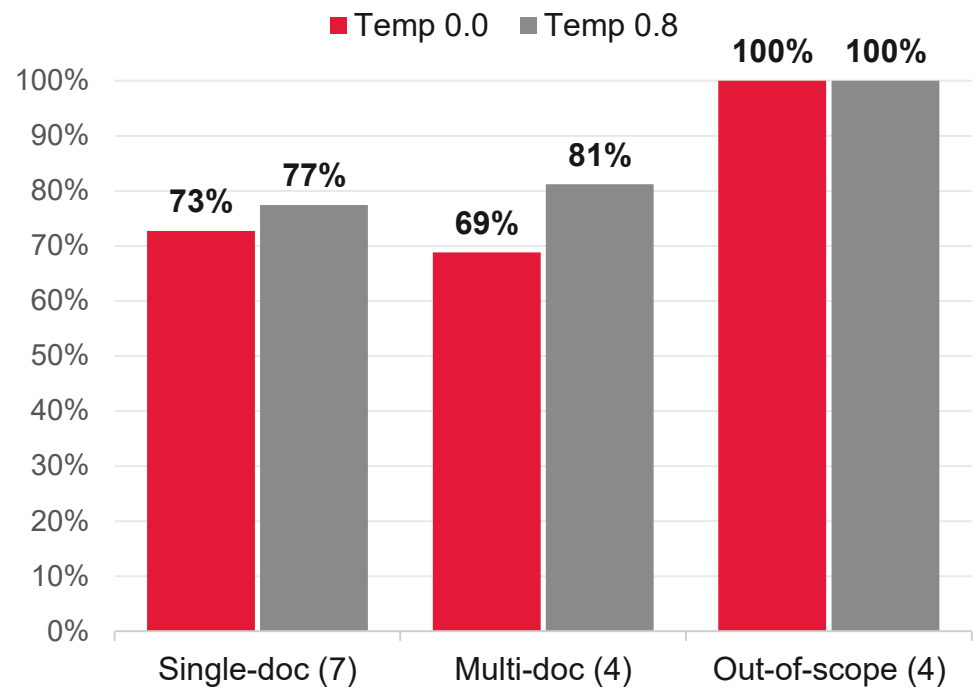
BENCHMARK RESULTS

No hallucinations flagged across 15 test questions

OVERALL (15 QUESTIONS)



CORRECTNESS BY QUESTION TYPE



0%
hallucination flagged, both configs

4 / 4
out-of-scope questions declined

~2s
avg latency per answer (approx.)

No clear gap
temp 0.0 vs 0.8, 15 questions

HALLUCINATION & RISK

RAG reduces hallucination risk. It does not remove it.

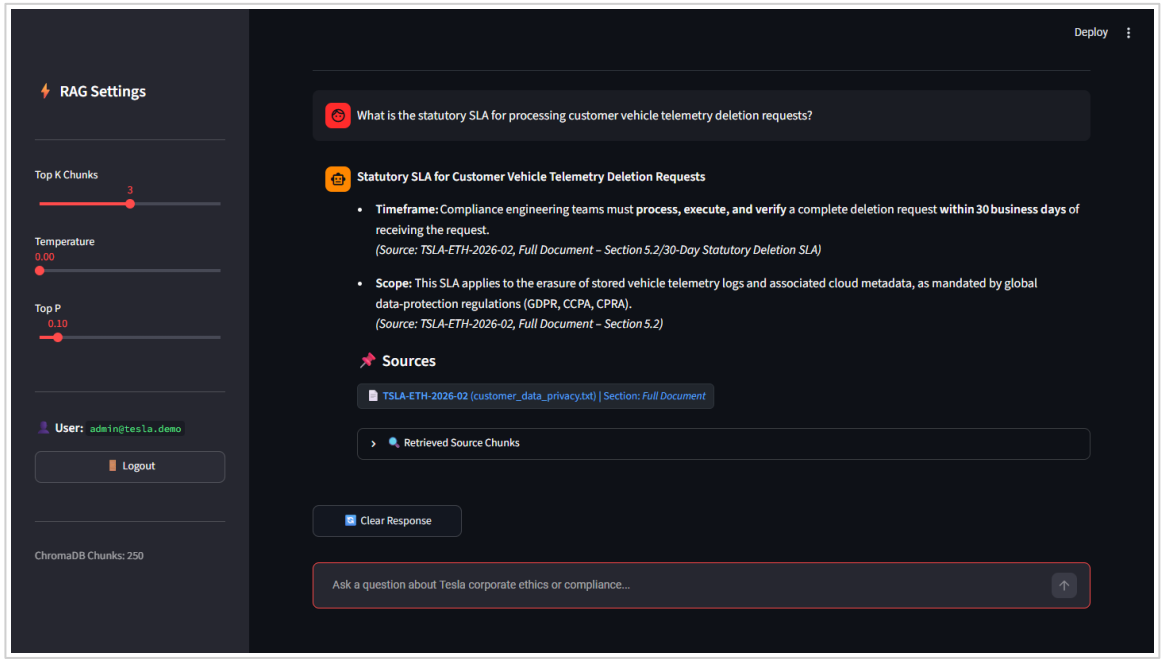
RISKS	IMPLEMENTED SAFEGUARDS	GAPS & PROPOSED NEXT STEPS
Misread policy A confident but wrong answer can be acted on.	**Evidence-only prompt** No outside knowledge, no invented rules.	**Section-level citations** Today: document level only.
Missed evidence 1 answerable multi-doc question was declined.	**Similarity cut-off 0.35** Weak matches are dropped.	**Clear error messages** API failures look like a config error.
False confidence Fluent text hides missing evidence.	**Decline when unsure** Fixed refusal sentence.	**Human escalation** Proposed: route to Compliance.
Cloud data flow Excerpts are sent to a hosted LLM.	**Visible sources** Doc ID badge plus excerpt inspector.	**Larger test set + review** Proposed: 100+ questions, expert scoring.

IMPLEMENTED

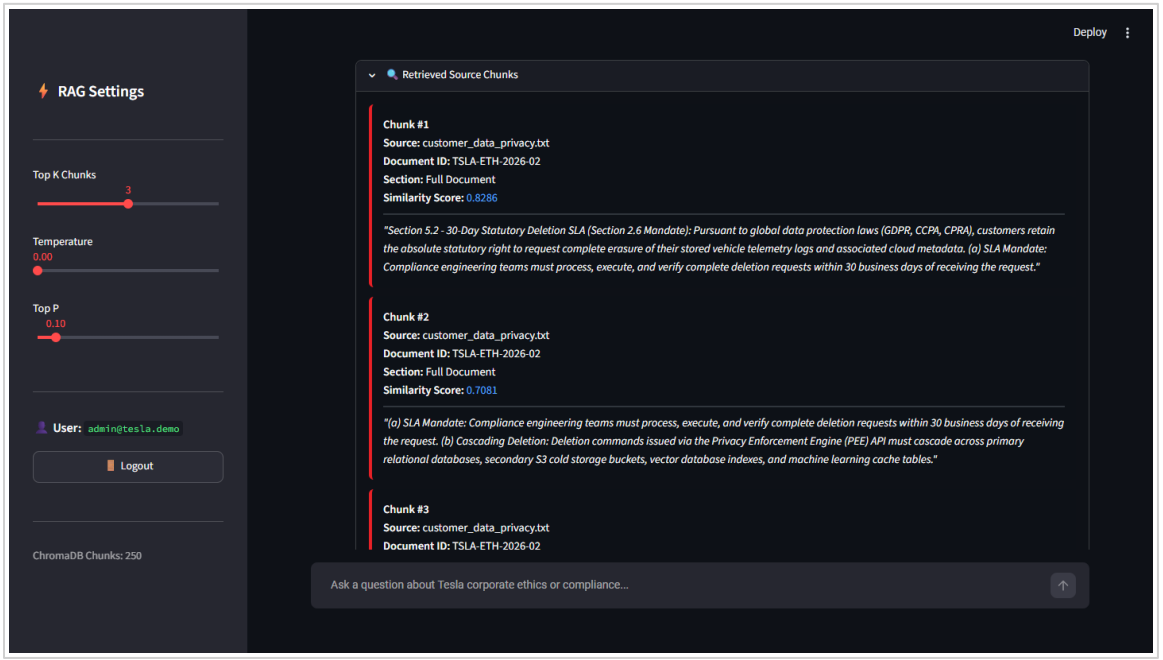
PROPOSED / FIXES NEEDED

APPLICATION SCREENSHOTS

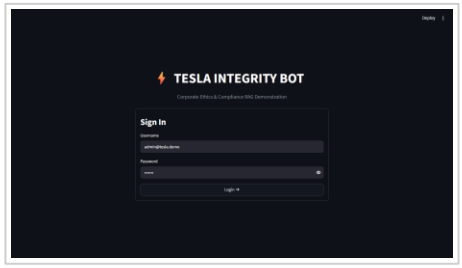
The working application



1 **Grounded answer with source badge**



2 **Retrieved chunks with similarity scores**



3 **Mock sign-in**

Unaltered captures of the running Streamlit app on the synthetic corpus. Any username is accepted at sign-in (mock gate). Sidebar exposes Top-K, temperature and top-p.

FUTURE SCOPE & RECOMMENDATION

Pilot first, then scale with controls

PHASE 1

Prepare

- Approved policy documents
- Section-level citations
- Clear error handling

PHASE 2

Pilot

- About 50 employees
- 100+ question test set
- Expert-scored accuracy

PHASE 3

Harden & scale

- SSO, role-based access, audit logs
- SLM vs. LLM comparison
- Human escalation path

ALL ROADMAP ITEMS ARE PROPOSED. NONE ARE IMPLEMENTED YET.

RECOMMENDATION

Approve a controlled pilot with approved policy documents. Measure accuracy and business impact, then add enterprise security before any wider rollout.